

Task 2: Servlet-Based CRUD Application

Project Component: Contact Manager Module

Core Description

Transform your static Personal Portal (Task 1) into a dynamic contact management system using Java EE technologies. This module enables:

- Visitor contact submissions
 - Admin management of contacts
 - Real-time form validation
 - Data persistence between sessions
-

Detailed Responsibilities

1. Servlet Development

- ☒ Create `ContactServlet.java` to handle:
 - GET `/contacts`: Display all contacts
 - GET `/contacts/add`: Show contact form
- ☒ Develop `AddContactServlet.java` to handle:
 - POST `/contacts`: Process form submissions
 - Validate input data
 - Redirect to success view or return errors

2. JSP View Design

- ☒ Build `contact-list.jsp`:
 - Display contacts in responsive table
 - Show "No contacts" placeholder when empty
- ☒ Create `contact-form.jsp`:
 - Form with fields: Name, Email, Phone
 - Real-time error display under fields

- Success/error message alerts

3. Validation & Error Handling

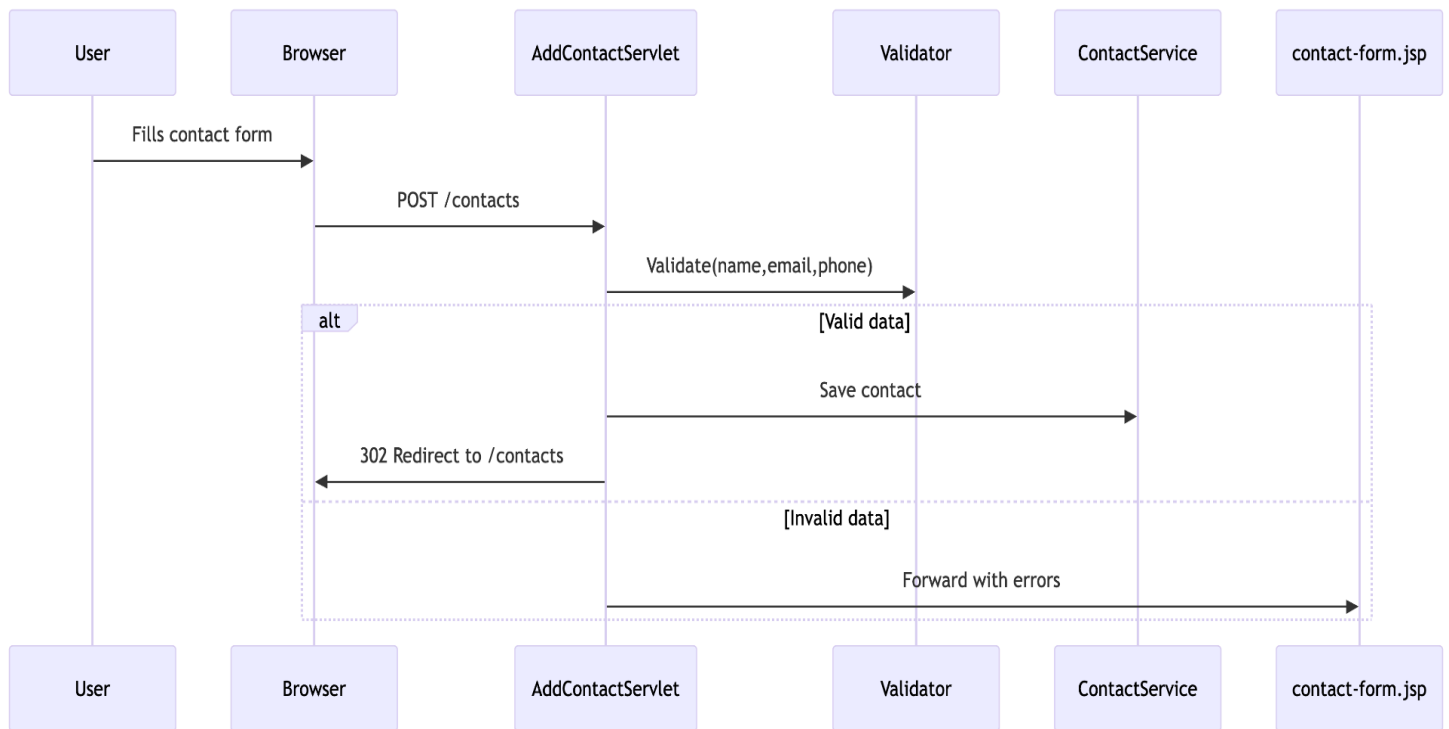
-  Implement server-side validation:
 - Required fields (name, email)
 - Email format verification
 - Phone number pattern matching
-  Design user-friendly error feedback:
 - Highlight invalid fields in red
 - Preserve valid form entries on error
 - Prevent duplicate submissions

Technical Specifications

Component	Technology	Responsibility
Controller	Java Servlets	Handle HTTP verbs (GET/POST), route requests
View Layer	JSP + JSTL	Render HTML with dynamic data
Data Model	JavaBeans	Store contact data (id, name, email, phone)
Validation	Java Utilities	Server-side input sanitization
State Management	HTTP Session	Maintain contact list between requests

Key Features to Implement

1. Contact Submission Workflow:
- 2.



3. Admin Interface:

- Contact list pagination
- Search-as-you-type functionality
- "Add New Contact" floating button

4. User Experience:

- Success toast notification on submission
- Auto-focus first invalid field
- Loading spinner during form processing

Validation Requirements

Field	Validation Rules	Error Message
Name	Required, 2-50 characters	"Please enter valid name"
Email	Required, valid email format	"Invalid email address"

Phone	Optional, 10-digit format if provided	"Use 10-digit format"
-------	---------------------------------------	-----------------------

Error Handling Responsibilities

1. Client-Side Errors (400 Bad Request):
 - Missing required fields
 - Malformed email addresses
2. Server Errors (500 Internal Error):
 - Handle `NullPointerException` in servlets
 - Gracefully manage session expiration
3. User Feedback:

```
<c:if test="${not empty error}">
  <div class="alert error">${error}</div>
```

```
4. </c:if>
```

Quality Assurance

1. Test Cases:
 - Submit empty form → Field-specific errors
 - Enter invalid email → "Invalid email" message
 - Valid submission → Contact appears in list
 2. Security Checks:
 - Prevent XSS via output escaping:
 - `<td><c:out value="${contact.name}" /></td>`
 - CSRF protection (for future implementation)
-

Expected Outcome

After completing this task, your Personal Portal will:

1. Have functional "Contacts" section at `/contacts`
2. Accept visitor submissions via web form

3. Display professional contact management interface
4. Maintain data integrity through validation
5. Provide intuitive user feedback

Note: Data is stored in-memory (will persist to database in Task 3). Focus on mastering Servlet/JSP workflow.